MAHATMA GANDHI UNIVERSITY KOTTAYAM

THIRD SEMESTER BCom (Honors)

Model Question Paper

DSC A Quantitative Techniques for Business (MG3DSCOM201)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Time 2 hours **Maximum Marks 70**

Section A

(Answer any 12 questions. Each question carries 1 Mark.)

1. Which is the best average for measuring the average speed of a Vehicle? (CO 1)
2. The graphical representation of a cumulative frequency distribution is known as ….(CO1)
3. What is a Lorenz Curve? (CO 2)
4. The mean of squared deviations about the mean is called …..(CO 2)
5. The first central moment is always …….(CO 3)
6. A distribution with skewness value of Zero is called ……(CO 3)
7. A matrix having only one column is called ……(CO 4)
8. The totality of all items under study is called ……(CO 5)
9. Calculate GM of the values 4 and 16. (CO 1)
10. The wages of 5 workers are Rs. 5,7,11,9,13 respectively. Calculate the variation by range method. (CO 2)
11. If the first quartile is 104 and the quartile deviation is 20, find out the value of upper quartile. (CO 2)
12. The matrix of order m x n indicates that there are ----rows and ---- columns. (CO 4)
13. Row headings in a table is also known as ……(CO 5)
14. Representative part of the population is called……(CO 5)

Section B

(Answer any 4 questions. Each question carries 7 Marks.)

15. Explain the various methods of collecting primary data. (CO 5)

16. How mode is located graphically? (CO 1)

17. The mean wage of 200 workers working in a factory is Rs. 50. The mean wage of 75 workers of the first shift is Rs.60. Find the mean wage of the rest. (CO 1)

18. The first of the two samples has 50 items with mean 7.5 and standard deviation 3. If the whole group has 150 items with mean 10 and standard deviation 3.5, find the standard deviation of the second group. (CO 2)

19. Let A = $\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$, B = $\begin{bmatrix} 3 & 2 \\ 5 & 2 \end{bmatrix}$, Find a) A + B, b) A -B, c) AB, d) BA. (CO 4)

20. For the following distribution calculate first four moments about mean. (CO 3)

Marks	5	10	15	20	25	30	35
Frequency	4	10	20	36	16	12	2

Section C

(Answer any 2questions. Each question carries 15 Marks.)

21. Find the mean, median and mode of the following data: (CO 1)

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	6	8	10	12	6	8

22. ABC Investment Company operates three mutual fund schemes: Equity Fund (E), Debt Fund (D), and Hybrid Fund (H). The company's portfolio allocation shows the percentage distribution of investments across three sectors: Technology (T), Healthcare (H), and Finance (F). Where:

Row 1: Equity Fund allocations (30% Technology, 25% Healthcare, 45% Finance)

Row 2: Debt Fund allocations (20% Technology, 40% Healthcare, 40% Finance)

Row 3: Hybrid Fund allocations (35% Technology, 30% Healthcare, 35% Finance)

(a) Prepare Portfolio allocation matrix A. (b) Calculate the determinant of matrix A

(c) Find the inverse of matrix A

(d) If the determinant is zero, what would this mean for the fund's investment strategy?

(CO 4)

23. Life of bulbs produced by two factories A and B are given below:

Length of life (in hours)	Factory A (Number of bulbs)	Factory B (Number of bulbs)
550 - 650	10	8
650 - 750	22	60
750 - 850	52	24
850 - 950	20	16
950 - 1050	16	12

The bulbs of which factory are more consistent from the point of view of length of life? (CO 2)